



**ABES Engineering College, Ghaziabad**  
Department of Applied Sciences & Humanities

Session: 2025-26

Semester: I

Course Code: 25AS102

Course Name: Fundamentals of Linear Algebra & Statistics

Max. Marks: 10

**Assignment 5**

Date of Assignment: 12/12/25

Date of submission: 18/12/25

| S.No.                        | KL    | CO    | Question                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks                        |       |       |     |     |    |     |    |       |        |       |       |       |       |     |     |    |   |   |
|------------------------------|-------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------|-------|-----|-----|----|-----|----|-------|--------|-------|-------|-------|-------|-----|-----|----|---|---|
| 1                            | K3    | CO5   | Find the least squares approximations of the form $y = ae^{bx}$ for the discrete data:<br><table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Y</td><td>7.209</td><td>5.265</td><td>3.846</td><td>2.809</td><td>2.052</td><td>1.499</td></tr></table>                                                                                                                                                       | X                            | 1     | 2     | 3   | 4   | 5  | 6   | Y  | 7.209 | 5.265  | 3.846 | 2.809 | 2.052 | 1.499 | 2   |     |    |   |   |
| X                            | 1     | 2     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4                            | 5     | 6     |     |     |    |     |    |       |        |       |       |       |       |     |     |    |   |   |
| Y                            | 7.209 | 5.265 | 3.846                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2.809                        | 2.052 | 1.499 |     |     |    |     |    |       |        |       |       |       |       |     |     |    |   |   |
| 2.                           | K3    | CO5   | The two regression equations of the variables x and y are $x=19.13-0.87y$ and $y= 11.64-0.50x$ . Find (i) mean of x & y (ii) correlation coefficient between x & y (iii) S.D. of y if S.D. of x=5.                                                                                                                                                                                                                                                    | 2                            |       |       |     |     |    |     |    |       |        |       |       |       |       |     |     |    |   |   |
| 3.                           | K3    | CO5   | Given the bivariate data:<br><table><tr><td>X:</td><td>1</td><td>5</td><td>3</td><td>2</td><td>1</td><td>1</td><td>7</td><td>3</td></tr><tr><td>Y:</td><td>6</td><td>1</td><td>0</td><td>0</td><td>1</td><td>2</td><td>1</td><td>5</td></tr></table><br>(i) Fit a regression line of y on x and predict y when x = 5<br>(ii) Fit a regression line of x on y and predict y when y = 2.5<br>(iii) Calculate Karl Pearson’s coefficient of correlation. | X:                           | 1     | 5     | 3   | 2   | 1  | 1   | 7  | 3     | Y:     | 6     | 1     | 0     | 0     | 1   | 2   | 1  | 5 | 2 |
| X:                           | 1     | 5     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2                            | 1     | 1     | 7   | 3   |    |     |    |       |        |       |       |       |       |     |     |    |   |   |
| Y:                           | 6     | 1     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0                            | 1     | 2     | 1   | 5   |    |     |    |       |        |       |       |       |       |     |     |    |   |   |
| 4                            | K3    | CO5   | Q.3. Calculate the rank correlation coefficient from the following data:<br><table><tr><td>Expenditure on advertisement</td><td>10</td><td>15</td><td>14</td><td>25</td><td>14</td><td>14</td><td>20</td><td>22</td></tr><tr><td>Profit</td><td>6</td><td>25</td><td>12</td><td>18</td><td>25</td><td>40</td><td>10</td><td>7</td></tr></table>                                                                                                       | Expenditure on advertisement | 10    | 15    | 14  | 25  | 14 | 14  | 20 | 22    | Profit | 6     | 25    | 12    | 18    | 25  | 40  | 10 | 7 | 2 |
| Expenditure on advertisement | 10    | 15    | 14                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 25                           | 14    | 14    | 20  | 22  |    |     |    |       |        |       |       |       |       |     |     |    |   |   |
| Profit                       | 6     | 25    | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 18                           | 25    | 40    | 10  | 7   |    |     |    |       |        |       |       |       |       |     |     |    |   |   |
| 5                            | K3    | CO5   | Fit a parabolic curve of regression of y on x to the following data:<br><table><tr><td>X</td><td>1</td><td>1.5</td><td>2</td><td>2.5</td><td>3</td><td>3.5</td><td>4</td></tr><tr><td>Y</td><td>1.1</td><td>1.3</td><td>1.6</td><td>2.0</td><td>2.7</td><td>3.4</td><td>4.1</td></tr></table>                                                                                                                                                         | X                            | 1     | 1.5   | 2   | 2.5 | 3  | 3.5 | 4  | Y     | 1.1    | 1.3   | 1.6   | 2.0   | 2.7   | 3.4 | 4.1 | 2  |   |   |
| X                            | 1     | 1.5   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2.5                          | 3     | 3.5   | 4   |     |    |     |    |       |        |       |       |       |       |     |     |    |   |   |
| Y                            | 1.1   | 1.3   | 1.6                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2.0                          | 2.7   | 3.4   | 4.1 |     |    |     |    |       |        |       |       |       |       |     |     |    |   |   |

**Answers:**

- $9.8632e^{-0.3141x}$
- 15.935, 3.67, -0.659, 6.5954
- (i)  $y = -0.304x + 2.874$ , 1.354 (ii)  $x = -0.2857y + 3.4464$ , 2.73215 (iii) 0.29480
- 0.024
- $y = 1.0357 - 0.19286x + 0.24286x^2$ .